Ashim Khanal, Ph.D.

Industrial and Management Systems Engineering, University of South Florida, Tampa, FL, 33620

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Teaching Interests

- Multi-objective Optimization Under Uncertainty
- Sustainable Supply Chain and Return Logistics Management
- Predictive & Prescriptive Analytics with Visualization
- Production and Operations Management
- Decision Making with Applied Deep Reinforcement Learning
- Engineering Economics and Financial Engineering
- Production Warehousing and Distribution Operations Management

Research Interests

- Novel Multi-objective Optimization Methods & Efficient Algorithms
- Smart Decomposition Methods for Large Scale Multi-Objective Network Optimization
- Sustainable Supply Chain, Circular Production Economy and Return Logistics
- Applied Deep Reinforcement Learning and Attention Networks for Production & Warehousing Management
- Sustainable Socio-Economic, Environmental and Critical Infrastructure Systems

Education

AUG 2020 – CURRENT (Expected May 2026)

Ph.D. Candidate (ABD) Industrial Engineering | University of South Florida | Tampa, FL

Potential Dissertation Topic: Multi-objective Optimization for Sustainable Systems: Methods, Algorithms and Application. Advisor: Dr Hadi Gard (Charkhgard)

Notable Coursework: Multi-objective Optimization, Integer Programming, Data Driven Optimization, Decision Making by Deep Reinforcement Learning

AUG 2020 – MAY 2023

M.S. Industrial Engineering | University of South Florida | Tampa, FL

Project: Randomized Adaptive Search Metaheuristics for Travelling Salesman and Vehicle Routing Problems

Notable Coursework: Network Modeling Design and Optimization, Data Mining, Statistical Fundamentals of Data Intelligence, Optimization Methods with Applications

SEP 2014 – JAN 2019

B.E. Industrial Engineering | Tribhuvan University | Kathmandu, Nepal

Seminar Project: Total Productive Maintenance (TPM) Practices and Implementation in a COKE bottling plant for Key Performance Indicators (KPIs) Improvement

Notable Coursework: Supply Chain Engineering and Management, Industrial Management, Quality Management, Micro and Macro Economics, Project Management, Operations Research

Experience

JAN 2021 – PRESENT

Graduate Research Assistant | Multi-Objective Optimization Lab | University of South Florida

- Innovated a novel method for solving large scale maximum multiplicative programs achieving 3 times faster computation with a maximum of 3% gap compared to state-of-the-art. Applications to Game Theory, Sustainability, Equity, Reliability

- Developed ‘AquaNutriOpt’, an open-source multi-objective, multi-period network optimization model for nutrient management in waterbodies as a Python package (available on PyPI); Funded by the Environmental Protection Agency
- Developed a generalized web-based watershed management system for Harmful Algal Bloom control by integrating automated hydrological model classification, AquaNutriOpt, and GIS visualization in a unified GUI; Funded by U.S. Army Corps of Engineers

JAN 2021 – PRESENT

Graduate Teaching Assistant | Industrial & Management Systems Engineering | University of South Florida

- Courses Assisted:
  • EIN 6935 Python for Data Science (master’s) (1 semester)
  • ESI 6410 Optimization Methods with Applications (master’s) (1 semester)
  • EGN 3615 Engineering Economic Analysis (bachelor’s) (7 semesters)
- Mentored both Graduate and Undergraduate level courses for over 300 students each year
- Duties involved teaching selected chapters by professors, leading recitation practice classes, grading exams, and helping students solve their course and project related queries in office hours. TA Duty Supervisor: Dr Jamie Chilton

NOV 2019 – MAR 2020

Assistant Lecturer – Part Time | Industrial Engineering | Tribhuvan University

- IE 505 Material Science & Metallurgy (bachelor’s) (2 classes; 1 semester)
- Delivered lectures and practical lab sessions for two engineering undergraduate classes of 48 students each, focusing on sustainable and environmentally friendly materials for engineering design
- Designed updated lab experiment tutorial syllabus according to the course purpose and learning goals

MAY 2019 – AUG 2019

Supply Chain Intern| Unilever Nepal Limited | Hetauda, Nepal

Project: Lean production planning and process improvement

- Reduced production changeovers from an average total of 117 to 105 per month, in four production lines combined. Techniques used: Mixed-Integer Programming & Data Analytics, Statistical Forecasting and Inventory Optimization

MAY 2018 – SEP 2018

Intern at Supply Chain Function | Bottlers Nepal Limited (COKE bottling) | Kathmandu, Nepal

Project: Implementation of total productive maintenance for production KPIs improvement

- Improved closure yield from 98.7% to 99.5% of a Fanta production line, enhancing production efficiency and reducing waste. Techniques used: AutoCAD mapping of lubrication points, Total Productive Maintenance and Management (TPM), and Root Cause Analysis (RCA)

Peer Reviewed Publications

1. **Khanal, A.,** Charkhgard, H., (2026) “Equitable Nutrient Management in Watersheds: A Game-Theoretic Optimization Framework to Control Harmful Algal Blooms” *IISE Annual Conference*
2. **Khanal, A.**, Charkhgard, H., (2025) “A Criterion Space Search Feasibility Pump Heuristic for Solving Maximum Multiplicative Programs” *Discrete Optimization* *Link*
3. **Khanal, A.**, et. al., (2024) “AquaNutriOpt: Optimizing nutrients for controlling harmful algal blooms in Python—A case study of Lake Okeechobee” *Environmental Modelling and Software* *Link*
4. **Khanal, A.**, et. al., (2024) “AquaNutriOpt II: A Multi-Period Bi-Objective Nutrient Optimization Python Tool for Controlling Harmful Algal Blooms - A Case Study of Lake Okeechobee” *Environmental Modelling and Software* *Link*
5. Tarabih, O.M., Arias, M.E., **Khanal, A.**, et al. (2024) “Effects of the spatial distribution of best management practices for watershed-wide nutrient load reduction” *Ecological Engineering* *Link*
6. **Khanal, A.**, et. al. “Generalized Approach to Integrated Modeling of Watershed, Water Body, and Best Management Practices Optimization with Graphical User Interface” Under Review ResearchGate *Link*
   Google Scholar Link

Conference Presentations

- Multi-Objective Optimization for Sustainable Watershed Management: Algorithms, Tools, and Decision Support System 2025 INFORMS Annual Meeting, Atlanta *(Invited)*
- Poster Competition: An Integrated Web Tool for Nutrient Optimization and Watershed Model Selection in Harmful Algal Bloom Mitigation, 2025 INFORMS Annual Meeting, Atlanta *(Third Place Winner)*
- Decision Support for Watershed Nutrient Management, Tethys Summit 2025, Tampa *(Plenary Talk)* Link
- AquaNutriOpt V2.0: A Bi-Objective Multi-Period Optimization Tool for Controlling Harmful Algal Blooms (HABs). INFORMS Annual Meeting 2024, Seattle *(Invited Session)* Link
- A Novel Optimization Model with Python Package for Controlling Harmful Algal Blooms in Lake Okeechobee. INFORMS Annual Meeting 2023, Phoenix *(Invited Session)* Link
- A Feasibility Pump based heuristic approach for maximum multiplicative integer linear programs. INFORMS Annual Meeting 2022, Indianapolis *(Invited Session)* Link

Upcoming Conference Paper Presentation

- Equitable Nutrient Management in Watersheds: A Game Theoretic Optimization Framework to Combat Harmful Algal Blooms. *2026 IISE Annual Conference, Arlington Texas*

Notable Projects

- **Network Modeling, Design, and Optimization:** Randomized Adaptive search Metaheuristics for solving Large-Scale Travelling Salesman Problem.
  Outcome: Maximum gap of 6.04% on networks involving over 1000 cities
- **Data-Driven Optimization:** Robust Network Optimization Modeling and Validation for Harmful Algal Bloom Control in Lake Okeechobee.
  Outcome: Robust optimization model reduced risk by limiting worst-case nutrient flow increases to ~5% compared to deterministic models, which showed up to 25% performance degradation under adverse conditions
- **Deep Learning:** Does Convolutional Neural Network trained on normal images recognize horizontally and vertically flipped images?
  Outcome: CNN recognized flipped images while trained on normal images for 10 % of the case. RESNET-50 architecture improved accuracy for such prediction to 65 %.
- **Deep Reinforcement Learning:** Solving Bi-Objective Network Optimization Model for lake-watershed management using Dueling DQN architecture
  Outcome: Achieved 5% better nutrient load estimation at target node than classical optimization heuristics.
- **Advanced Engineering Analytics:** Multi-class Cyber Malware classification based on API calls on a server using advanced analytics models with data wrangling
  Outcome: Synthetic data generation for class balancing and principal component analysis improved the accuracy of random forest classifier and neural network classifier from 25 % to 90 %
- **Integer Programming:** Improving the Feasibility Pump Heuristic for Mixed Integer Programs
  Outcome: 5-fold cross validation and multi-layer perceptron regressor model predicted the number of variables to be flipped and Perturbation Frequency to avoid stalling issues

Skills

Operations Research & Optimization: Mixed-Integer Linear Programming, Multi-Objective Optimization, Large Scale Decomposition Methods, Convex Optimization, Robust Optimization, Stochastic Programming, Combinatorial Network Optimization, Gurobi, CPLEX, SCIP, Coin OR CBC, Ipopt

Engineering Management: Supply Chain & Logistics Management, Total Productive Management, Total Quality Management (TQM), Economics & Financial Analysis, Lean Manufacturing, Operations Management, Root Cause Analysis, Game Theory &

	Strategy, Smart & Sustainable Management System Design, Descriptive, Prescriptive, and Predictive Analytics
Machine Learning & Analytics	TensorFlow, PyTorch, Scikit-Learn, Pandas, Matplotlib, Seaborn, Spark, Weka, Data Preprocessing and Wrangling, Reinforcement Learning, Deep Learning Models, Deep Reinforcement Learning Algorithms
Programming	Python, C++, Julia, R, SQL, Shell Script, Data Structures & Algorithms, Excel VBA
Collaborative Development	Git, Ubuntu, Docker, VS Code, PyCharm, Latex
Soft Skills	Leadership, Innovativeness, Time-Management, Teamwork, Problem-Solving, Effective Communication, Analytical Thinking, Self-Motivated, Solution Oriented, Conflict Management, Flexibility, Adaptability

Leadership and Volunteering

President - INFORMS Student Chapter at University of South Florida 2023-24

- Winner *Summa Cum Laude* Best Student Chapter Award at INFORMS Annual Meeting 2024, Seattle.
- Organized 11 faculty lecture series, hosting faculty scholars from prestigious US universities
- Organized an international data science boot camp collaborating with universities in Peru, Nepal, and Chile, attracting 45 participants
- Reference: Dr Jose Zayas-Castro (Chapter Advisor & Mentor) *Email*

President - Nepalese Student's Association at University of South Florida 2022-23

- Organized cultural events engaging 200+ students
- Launched sustainable furniture donation program for new incoming Nepalese students
- Boosted association funding by 60% through strategic event planning, outreach, and attendance improvements

Student Volunteer at 17th INFORMS COMPUTING Society CONFERENCE, Tampa, 2022

- Managed registration desks and coordinated logistics for 90+ attendees, streamlining check-in processes.
- Reference: Dr. Hadi Charkhgard (Co-chair of the conference) *Email*

Student Volunteer – INFORMS Annual Meeting 2021, Anaheim

- Assisted presenters and attendees with virtual session navigation and technical support enhancing and easing virtual (COVID 19) experience

Awards

Third Place Winner	Poster Competition at 2025 INFORMS Annual Meeting, Atlanta *Link*
Summa Cum Laude	Best Student Chapter Award as a President of INFORMS Student Chapter at USF at 2024 INFORMS Annual Meeting, Seattle *Link*
Semester Topper	Highest GPA in the class of 2018 for three semesters and third rank in overall
Merit Scholarship	Full Tuition Scholarship for 4-year undergraduate degree, competing with over 15000 examinees, based upon the nationwide Institute of Engineering (IOE) entrance examination
Student of the Month	Best performance in academics (GPA) among high school colleagues

Professional Affiliation

- Member, Institute for Operations Research & Management Sciences (INFORMS)
- Member, Institute of Industrial and Systems Engineers (IISE)
- Member, Tethys Geo Science Foundation

Certifications

- Data Science and AI, Coursera 2020
- Microsoft Office Specialist Certification, 2017
- Energy Efficiency and Management, Government of Nepal, 2017

Languages

- Nepali (Native), English (Proficient)

References

Hadi Gard (Charkhgard) | Associate Professor, Department of Industrial & Management Systems Engineering, University of South Florida, Tampa, FL | Phone: +1 404 7365080 | hcharkhgard@usf.edu | *Major Ph.D. Advisor*

Jamie Chilton | Associate Professor of Instruction| Department of Industrial & Management Systems Engineering, University of South Florida, Tampa, FL | Phone: +1 8139747918| jmchilton@usf.edu *Teaching Assistant Supervisor*

Mauricio E. Arias | Associate Professor, Department of Civil & Environmental Engineering, University of South Florida, Tampa, FL | Phone: (813) 974-5593 | mearias@usf.edu *Dissertation Committee Member*

Ankit Shah | Assistant Professor | Department of Operations & Decision Technologies, Kelly School of Business, Indiana University - Bloomington, IN | Phone: 812-855-6331 | ankit@iu.edu | *Business School Professional, Mentor Reference*

Jose Zayas-Castro | Professor, Department of Industrial and Management Systems Engineering, University of South Florida, Tampa, FL | Phone: +1 813-789-1125 | josezaya@usf.edu | *Leadership Mentor, INFORMS Student Chapter Advisor*